SPARKLETEAM: Towards Recursive Self Home Improvement

Lifting a weak open model at dishwasher loading with a self-rewriting team

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Abstract

How best to load a dishwasher is a combinatorial, physical problem with an unusually honest reward: a simulated wash decides which items come out clean [8]. We use it as a testbed for SPARKLETEAM, a self-improving multi-agent system built only from open, inexpensive models. A deliberately weak loader (Nemotron-3.5-Super-VL, thinking off) is surrounded by a planner, a vision critic, a coach and a librarian, all open Nemotron models on Crusoe serverless inference. Between generations the team rewrites a typed genome of its own prompts, playbook, tool policy and structure, and a deterministic gatekeeper keeps an edit only if it beats the incumbent on paired, fresh tasks. On a held-out exam whose levels contain item kinds and soils never seen in training, the evolved team scores 63.7% against 46.0% for the bare loader (+17.8 pp). We report what changed, what it cost, and why the evidence is thin.

1 Introduction

Recursive self-improvement is usually discussed for systems that rewrite their own weights. A humbler, testable version keeps the model fixed and lets the system rewrite the *scaffold* around it: prompts, memory, tools and the division of labour [20, 65, 67]. Such loops are hard to trust: the reward is often an LLM judge, which can be flattered [70], a loop optimising a proxy finds its seams [27, 33, 52], and gains are usually measured where they were selected.

We take the problem of loading a dishwasher, using DishwasherBench, Composo's open (MIT) benchmark [8]. It models one real machine (Bosch SMV4HAX40G): items are placed with deterministic rigid-body physics (Jolt) through 13 MCP tools, then a GPU wash simulation (WebGPU via Dawn) of the Eco 50 programme with one detergent capsule, calibrated on washes of the real machine (parameter set `anchored-v8.owner-capsule`), decides which items are clean against a threshold derived from the IEC 60436 scale [6, 12, 21]. An item scores its points only if it comes out clean. The reward is *objective* (points over points available), *deterministic* (a load always settles and washes the same way, and every episode is verifiable by replay) and *unfakeable*: a bowl mouth-up under a dead spot of the spray arm stays dirty however eloquently the agent describes it. It is also hard: not everything fits, so the player chooses what to load as well as how; jets reach only part of the load [43, 44]; and choosing a subset under capacity is a knapsack over strongly NP-hard 3D packing [24, 34]. Household benchmarks score symbolic goals [45, 49, 51, 55], and prior dishwasher-loading work scores grasp success or agreement with a human preference [22, 23, 30, 56]; to our knowledge none scores whether the items come out clean, the gap DishwasherBench fills.

The benchmark, simulator, calibration, levels and harness are Composo's work [8]. Ours is SPARKLETEAM, the self-improving team, with its memory, Gatekeeper, exam protocol and visualiser (Fig. 1), plus cosmetic water, watch and turntable changes to the benchmark's web client that touch neither simulation nor scoring. We claim that a team of cheap open models can lift a weak loader by rewriting the system around it, keeping only changes that measurably help, and that the lift survives on unseen tasks. Contributions:

- SPARKLETEAM, a team whose rewritable state is a typed, versioned *genome*, edited only through six typed operations, with the judging roles (memory accounting, acceptance) in deterministic code: the team can change how it plays, never how it is graded.
- A paired acceptance rule over shared seeds in a deterministic simulator, a three-layer memory with shrunken Bayesian confidence, and a held-out exam on task families with new kinds and soils.
- An end-to-end run on open Nemotron models, with lineage, costs and limits.

2 Related Work

Scaffold self-improvement. STOP recursively improves the improver around a fixed LM [65]; ADAS's meta-agent programs new agents [20]; the Darwin Gödel Machine keeps self-edits that validate, in an open-ended archive [67]; Gödel Agent and SICA rewrite their own logic or code [47, 63]. AFlow searches workflows with MCTS and lets cheap models match expensive ones [66]; GPTSwarm optimises prompts and agent connectivity [71]; FunSearch and AlphaEvolve gate LLM proposals with automatic evaluators [36, 48]. Our Coach is a typed, constrained meta-agent in this line.

Prompt and program optimisation. OPRO optimises from a scored history [62]; Promptbreeder and EvoPrompt evolve prompts by LLM mutation [13, 16]; DSPy and MIPRO tune multi-stage LM programs against a metric [25, 40]; TextGrad and Trace propagate textual feedback [7, 64]; GEPA's reflection on traces is closest to our Critic→Coach loop [1].

Memory and multi-agent systems. Reflexion stores verbal reflections [50]; ExpeL and CLIN distil and transfer cross-task insights [32, 68]; Voyager and AWM grow verified skill or workflow libraries [59, 60]; see also [41, 42]. Unlike self-refinement [31], our lessons are counted against physics. Role-based teams [19, 28, 46, 61] and debate [10] follow fixed processes; ours edits its own structure. Voting and verified repeated sampling trade compute for parameters [4, 29, 53]; our verifier is too slow to call per action.

Search over loads. Loading is constrained packing with a physical, black-box objective [58, 69]; genetic algorithms [15, 18], CMA-ES over poses [17], quality-diversity [35], Bayesian optimisation [54] and MCTS over placement sequences [5, 26] are natural per-instance optimisers. We do not search loads directly: every evaluation costs a wash, and the product is *one load*, not a better loader, so nothing transfers to a new set of items. We optimise the system and test whether its improvements transfer.

3 Method

The team (Table 1, Fig. 2). In SPARKLETEAM the Loader plays through the standard DishwasherBench harness tools [8], with renders as at most one image per request (the newest). Optionally a Planner turns the level card and a memory brief into a per-item plan (rack, stance, what to leave out). After each practice episode the Critic gets the item-by-item outcome (rack, stance, clean, residue, retained water), plan, playbook and two renders of the final load, and writes up to five

*Built at the Crusoe × Cerebral Valley Recursive Agents Hackathon, 17 October 2026. Code: `https://github.com/elevenarms/RSHI`.

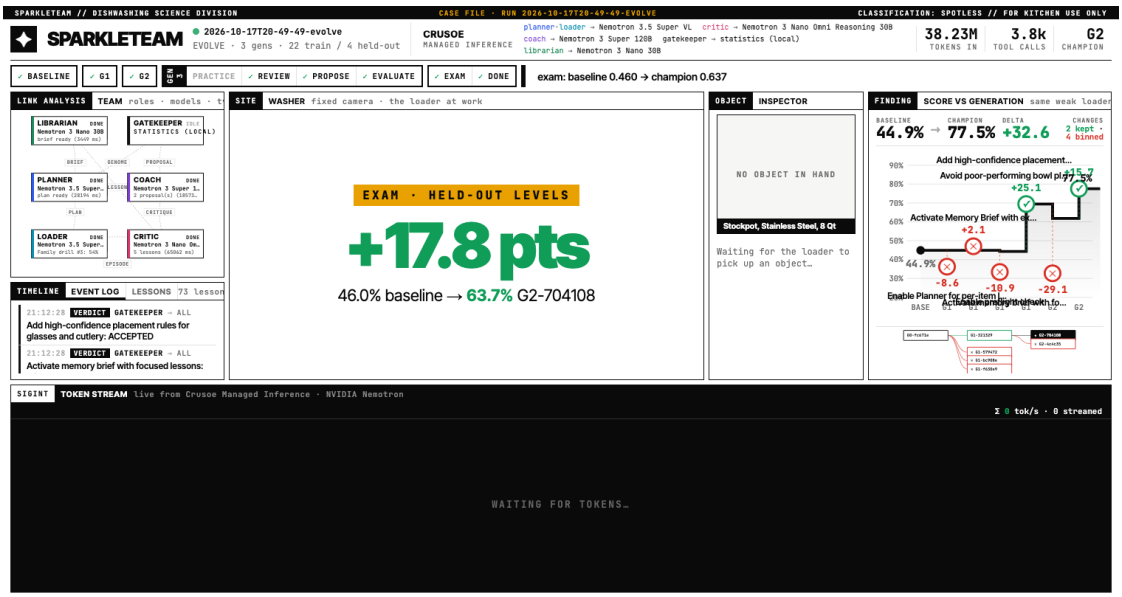


Figure 1: The Studio, the live visualizer of SPARKLETEAM: the team as a link-analysis graph with each role's Nemotron model; the timeline with Gatekeeper verdicts; the SITE panel, which shows the loader at work in the DishwasherBench 3D simulation during play and the exam finding at the end (46.0% → 63.7%); score vs. generation with accepted (green) and rejected (red) proposals and the genome lineage, whose percentages are *validation*-drill means per generation, not exam scores; and the token stream panel, which shows every token from Crusoe live during play.

Role	Model (open, on Crusoe)	Job
Loader	Nemotron-3.5-Super-VL (no thinking)	plays, 13 tools
Planner	Nemotron-3.5-Super-VL (low effort)	per-item plan
Critic	Nano-Omni-Reasoning-30B (thinking)	post-mortem
Coach	Nemotron-3-Super-120B-A12B	genome edits
Librarian	Nemotron-3-Nano-30B-A3B + code	memory, brief
Gatekeeper	deterministic statistics	accept/reject

Table 1: Roles and models [37–39].

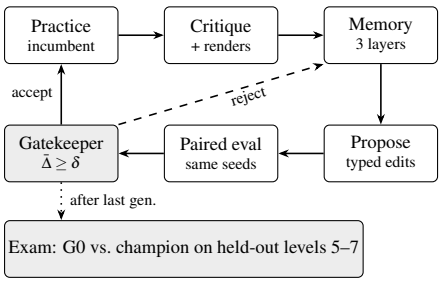


Figure 2: One generation of the loop. Rejections (title, hypothesis, per-task deltas) are logged and shown to the Coach.

tagged lessons with a polarity. The Coach proposes edits; the Librarian condenses memory into a brief.

Genome and typed edits. A genome is an immutable record of everything the team may rewrite: three prompts (Planner, loader preamble, Critic); a playbook of at most 25 tagged rules with evidence links to lessons; team switches (`usePlanner`, `useMemoryBrief`, `preflightCheck`, which refuses `start_cycle` before a read `check_load`, and the brief's lesson count); and loader settings (reasoning effort, images, denied tools, turn limit). The Coach can change it only by `add_rule`, `edit_rule`, `remove_rule`, `set_prompt`, `set_team` and `set_loader`. A proposal carries a title and a falsifiable hypothesis, should change one coherent thing so gains can be attributed, and is shown past rejections. Invalid edits are dropped; ids hash content and parent, so lineage is diffable. The baseline G0 is the bare loader: no planner, no brief, no preflight, an empty playbook.

Three-layer memory. The *episodic* layer keeps every training episode item by item; the *semantic* layer keeps lessons (from the Critic, and observed from placement statistics), deduplicated on merge, with support s and contradiction c counts; the *procedural* layer keeps every genome and decision. Confidence is a Beta(1,1) posterior mean shrunk towards ignorance while evidence is thin:

$$p = \frac{s+1}{n+2}, \quad \kappa = \tfrac{1}{2} + \left(p - \tfrac{1}{2}\right)\frac{n}{n+K}, \quad n = s+c,\ K = 3, \qquad (1)$$

so one supporting episode gives 0.54, five of five 0.72, twenty of twenty 0.90. A level's brief lists smoothed clean rates $(k+1)/(m+2)$ of its kinds per rack and stance, then lessons with $\kappa \geq 0.5$ ranked by $\kappa \cdot \text{relevance} \cdot \log(1+n)$. Exam episodes are stored apart and never read back.

Gatekeeper. Candidate C and incumbent I play the same fresh validation drills with the same seeds. Let P be the drills both played validly and x_t the share on drill t; then

$$\bar{\Delta} = \frac{1}{|P|}\sum_{t \in P}\left(x_t^C - x_t^I\right), \qquad \text{accept} \iff \bar{\Delta} \geq \delta, \qquad (2)$$

and the decision is void if $|P| < 3$. Agent failures (turn or token limit, nothing loaded) score what they scored; *invalid* episodes (provider or harness errors) are replayed on the same task and seed up to twice, and drills still unpaired are dropped for both genomes.[1] Pairing matters because episodes vary enormously: the same baseline genome scored 88% on a Tutorial drill and 0% on a Weeknight drill in one evaluation. Since loading and washing are deterministic, pairing cancels drill difficulty, leaving the genome's effect plus sampling noise. The accepted candidate with the largest $\bar{\Delta}$ becomes the incumbent; its validation episodes are the next practice set.

Exam: what "held-out" means. Training and validation drills come from DishwasherBench course levels 1–4 (tutorial, lunch for four, weeknight dinner, breakfast), the exam from levels 5–7 (mixed family, baking day, dinner party) [8]. These are new task families, not just new seeds: they hold kinds and soils training never showed (baby bottle and teat, spouted beakers, lunch tubs, baking tray, tin, platter, board; burnt sugar, overnight soils). G0 and the champion play identical exam drills; nothing is learned from them.

[1] In a pilot, HTTP 429s from a shared endpoint were first scored as agent failures and generation 2 had to be voided: on cheap shared endpoints, an acceptance rule must not confuse infrastructure noise with policy quality.

Loader (thinking)	Share	Loaded/clean	Turns	Time
DeepSeek-V4.1-Flash (off)	100%	12/12	21	90 s
Qwen3.8-27B (off)	100%	12/12	19	199 s
GLM-5.3-Flash (low)	93%	10/10	40	97 s
DeepSeek-V4-Flash (off)	89%	12/11	40	89 s
gpt-oss-120b (low)	49%	8/7	40	154 s
Nemotron-3-Nano-30B (low)	41%	5/5	40	870 s
Nemotron-3.5-Super-VL (off)	**21%**	**3/3**	**40**	**129 s**
Nemotron-3-Nano-30B (off)	0%	1/0	loops	
Nemotron-3-Nano-Omni (off)	0%	2/0	loops	

Table 2: Choosing the loader: one smoke per model on the same 12-item Breakfast drill (seed 7, 61 points, 40-turn cap). We take the weakest that still plays.

Exam drill	Source	Avail.	G0	Champ.	Δ
Family #0	mixed-family	59	48	42	−10.2
Baking #1	baking-day	71	28	63	+49.3
Party #2	dinner-party	77	15	32	+22.1
Family #3	mixed-family	71	31	38	+9.9
Total points		278	122	175	
Mean share			46.0%	63.7%	+17.8

Table 3: Held-out exam: points per 12-item drill for G0 and the champion G2-704108, same drills and seeds, fast tier; Δ in pp of share.

4 Experimental Setup

All models run on Crusoe Managed (serverless) Inference via an OpenAI-compatible API, with a client-side rate limiter and byte-stable prompts for prefix caching (83–90% of input tokens hit the cache in smoke runs). In a sweep (40-request bursts, 60 s at 8 in flight) only Nemotron-3.5-Lightning and Gemma-4-31B returned HTTP 429s; the Nano models sustained ∼300 calls/min at 370 ms median.

Choosing a weak loader. A saturated loader leaves the Gatekeeper nothing to measure (Table 2). Nemotron-3.5-Super-VL without thinking is the weakest that still plays: all it loads is clean, but it runs out of turns. Without thinking the Nano models loop; with it, Nano-30B plays but takes ∼6× Super-VL's wall time, too slow for an evolution loop. Our pilot used Nemotron-3.5-Lightning-30B-A3B (41–70% in smokes) until its shared deployment throttled us to about five calls per minute.

Protocol. A drill is 12 items of a DishwasherBench level drawn by seed, cutlery in proportion. The final run (`2026-10-17T20-49-49-evolve`) has 3 generations: 4 baseline practice drills, 2 proposals and 3 paired fresh validation drills per generation, $\delta = 0.03$, a 40-turn cap per episode, 8 game servers with one wash queue each and 8 episodes in flight, then an exam on 4 held-out drills (Table 3). Washes use the fast tier (4 mm blocks rather than 2 mm), whose verdicts agree with the official tier's on 91% of the standard load and 97% of pieces on the course levels [8].

Baseline. G0 scored 39% in practice: Breakfast 56% (8 of 8 loaded items clean), Tutorial 44% (8 of 8, washed early), Weeknight 32% (3 of 5), Lunch 24% (2 of 2). It fails at committing within the turn budget, not at knowing what washes clean.

5 Results

Exam. The champion (G2-704108, accepted in generation 3) scores a mean share of 63.7% against 46.0% for G0 (+17.8 pp; 175 vs. 122 of 278 points), ahead on 3 of 4 drills (Table 3); neither side had an invalid episode. It loads 35 items against 23 (of 48), with clean-of-loaded rates of 91% (32/35) and 100% (23/23): as share = loaded fraction × clean rate predicts from the baseline's failure mode, the gain is commitment, bought with a few dirty items.

Evolution. The Gatekeeper accepted 2 of 6 candidates (Table 4), by +0.251 and +0.157; memory ended with 36 lessons from 39 episodes (plus 2 replays). The champion differs from G0 by three playbook rules and nothing else: R1.1 moves one cereal bowl from a tilted upper-rack stance to the lower rack on edge (or out); R2.1 "Glasses and tumblers: upper_rack, mouth_down"; R2.2 "Cutlery: cutlery_basket, upright", the last two restating top memory lessons ($\kappa = 0.69$). Every structural edit (planner, memory brief, preflight) was rejected.[2]

G	Proposal (Coach's title, abridged)	Inc.	Cand.	W/L	
1	Enable Planner	.449	.363	1/2	rej.
1	Memory Brief, expanded lessons	.449	.471	1/2	rej.*
2	Enable preflight check	.443	.334	1/2	rej.
2	Avoid poor-performing bowl placement	.443	.695	3/0	**acc.**
3	Memory brief, focused lessons	.618	.327	1/2	rej.
3	Rules for glasses & cutlery	.618	.775	2/1	**acc.**

Table 4: Gatekeeper decisions: mean validation share of incumbent and candidate on 3 paired drills (Tutorial, Lunch, Weeknight; fresh seeds each generation, so levels compare only within a generation). *Accepted at +0.129 under the corrected replay rule (footnote 2).

Cost. 22.7 min of evolution plus about 7 min of exam on 8 game servers. The loader made 1,481 requests (38.24M input tokens, 83.7% cached; 0.43M output); the other roles 21 requests and under 0.1M tokens. Inference cost: about $0.12 for the Critic, Coach and Librarian at listed prices; Super-VL's price is not listed (at Super-120B's rates the loader would cost about $7.7).

6 Discussion and Limitations

Physics scores, code decides, models only propose: the division of authority LLM-judged loops lack [2, 70]. For this loader, crisp rules distilled from memory beat extra scaffolding: the planner, brief and preflight all lost, and two short rules won. In the Lightning pilot's generation 1 the opposite held: the Planner was accepted at +0.135 and a bowl-mouth-down rule rejected at −0.095. The best scaffold appears to be loader-specific, which is exactly why it must be found by measurement rather than assumed. The champion lost one exam drill (−10.2 pp, Family #0).

Limitations. *Small n*: four exam drills with a single seed each, so +17.8 pp is a point estimate; a one-sided sign test on 3 wins of 4 gives $p = 5/16 \approx 0.31$ [9]. This is a demonstration, not a statistically established effect. *One machine*: one dishwasher, programme and detergent; the exam tests transfer to new items, soils and loads, not new physics. *Stochastic sampling*: pairing removes task variance, not policy variance. *Fast tier*: a few percent of verdicts differ. *Multiple comparisons*: keeping the best of several candidates biases validation gains upwards, and repeated accept/reject overfits a benchmark [3, 11, 14]; hence the held-out exam. *Margin*: $\delta = 0.03$ over 3 drills is about one average item on one drill, far below the noise of a sampling loader; a sequential test [57] or repeated validation would be the principled fix.

7 Conclusion

A team of open, inexpensive models can rewrite the system around a weak loader, keep only what the physics confirms, and carry the improvement to tasks with objects it has never handled. Its reward cannot be argued with, which is exactly what recursive self-(home-)improvement needs. Next: more exam passes and an official-tier exam.

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[2] G0's first generation-1 Weeknight attempt (0 of 87; a pasta pot blocked the door, so the wash was refused) was wrongly classed as invalid and replayed (to 0.322). Under the corrected rule, fixed after the run (agent-caused non-washes score 0; only provider or harness failures are replayed), the incumbent's mean is 0.342, the Memory Brief would have been accepted at +0.129 and the Planner rejected at +0.021. The bug made the Gatekeeper more conservative; the champion's lineage (generations 2 and 3) was not affected.

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